LAB ASSIGNMENT

Programming Fundamentals

Lab - 2

NAME: Nisha Fatima

CLASS: BSCS 1st D

ROLL NO: 3

SUBMITTED TO: Sir Asad Majeed

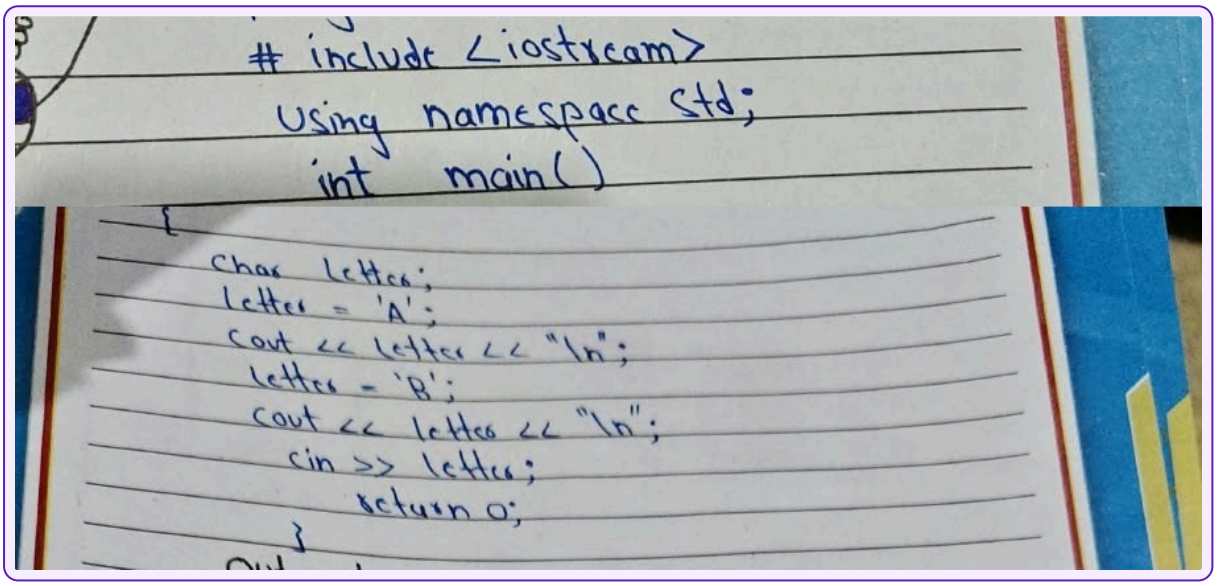
DATE: 11 October 2026

Lab Task 1:

Type and save the following programs in Visual Studio. Run these programs and observe their output.

Program 1:

HANDWRITTEN CODE



PHONE CODE

Coding C++
Auto saved at 10:05:14
RUN MENU

```cpp
#include<iostream>
using namespace std;
int main()
{char letter;
letter='A';
cout<<letter<<"\n";
letter='B';
cout<<letter<<"\n";
cin>>letter ;
return 0;
}

```

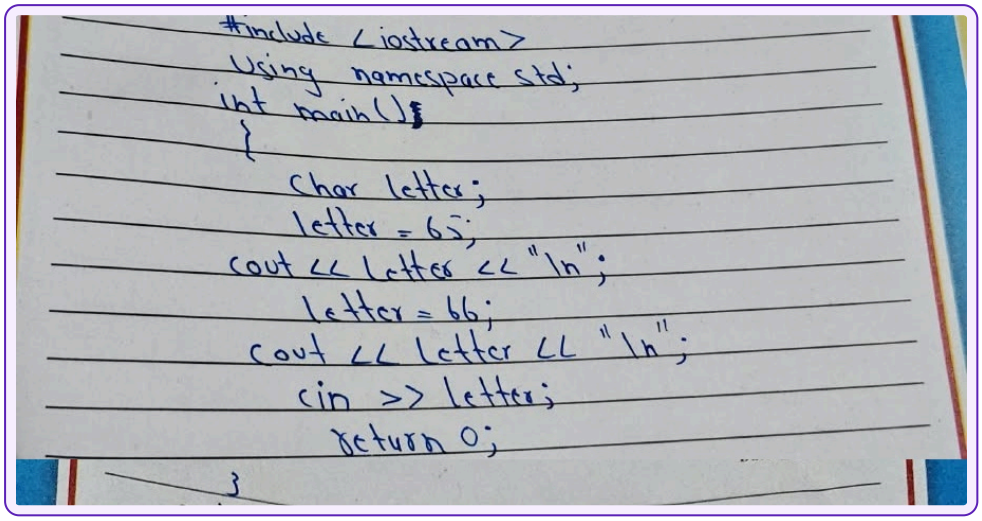
PHONE OUTPUT

Compile Result

```
A
B
```

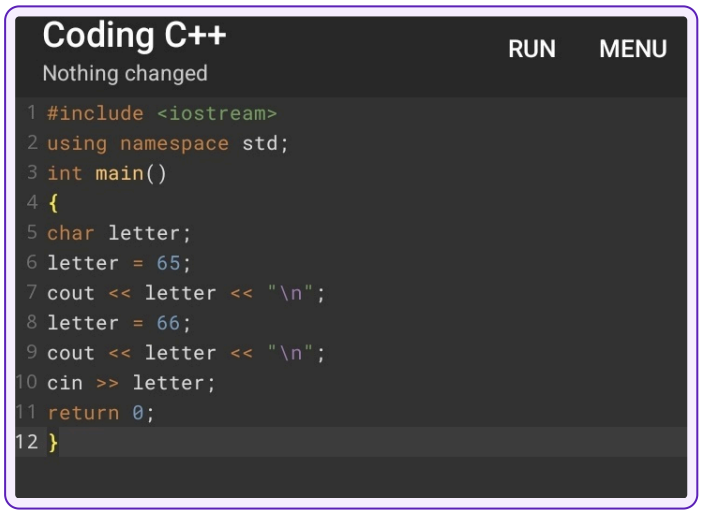
Program 2:

HANDWRITTEN CODE



```
#include <iostream>
Using namespace std;
int main(){
{
    Char letter;
    letter = 65;
cout << letter << "\n";
    letter = 66;
cout << letter << "\n";
    cin >> letter;
    return 0;
}
```

PHONE CODE



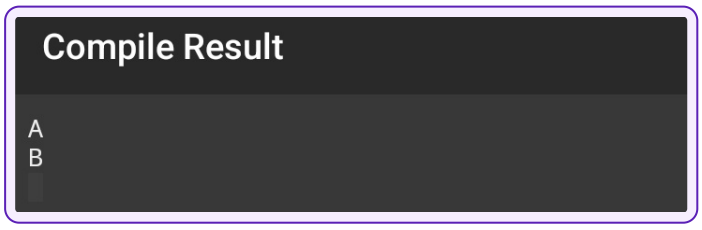
Coding C++

Nothing changed

RUN MENU

```cpp
#include <iostream>
using namespace std;
int main()
{
char letter;
letter = 65;
cout << letter << "\n";
letter = 66;
cout << letter << "\n";
cin >> letter;
return 0;
}
```

PHONE OUTPUT

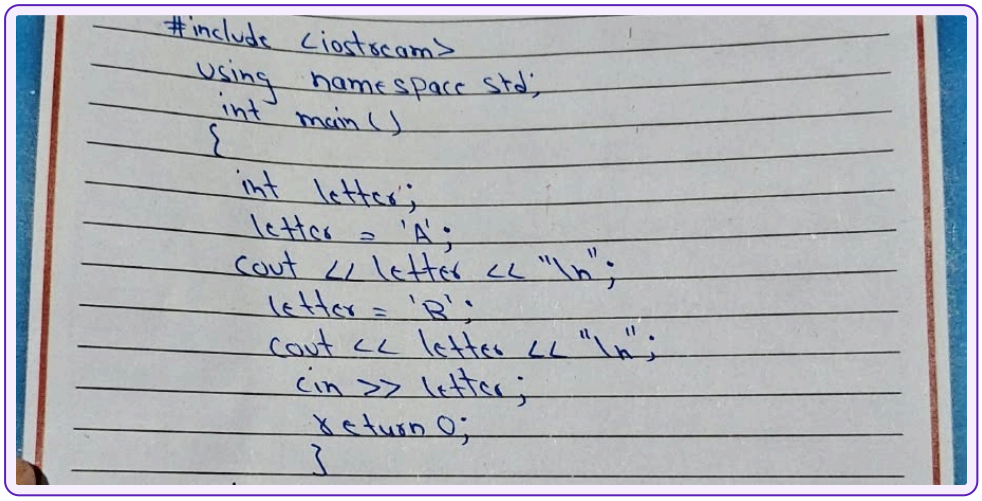


Compile Result

```
A
B
```

Program 3:

HANDWRITTEN CODE



```
#include <iostream>
using namespace std;
int main()
{
int letter;
letter = 'A';
cout << letter << "\n";
letter = 'B';
cout << letter << "\n";
cin >> letter;
return 0;
}
```

PHONE CODE

Coding C++
Nothing changed
RUN MENU

```
1 #include <iostream>
2 using namespace std;
3 int main()
4 {
5 int letter;
6 letter = 'A';
7 cout << letter << "\n";
8 letter = 'B';
9 cout << letter << "\n";
10 cin >> letter;
11 return 0;
12 }
```

PHONE OUTPUT

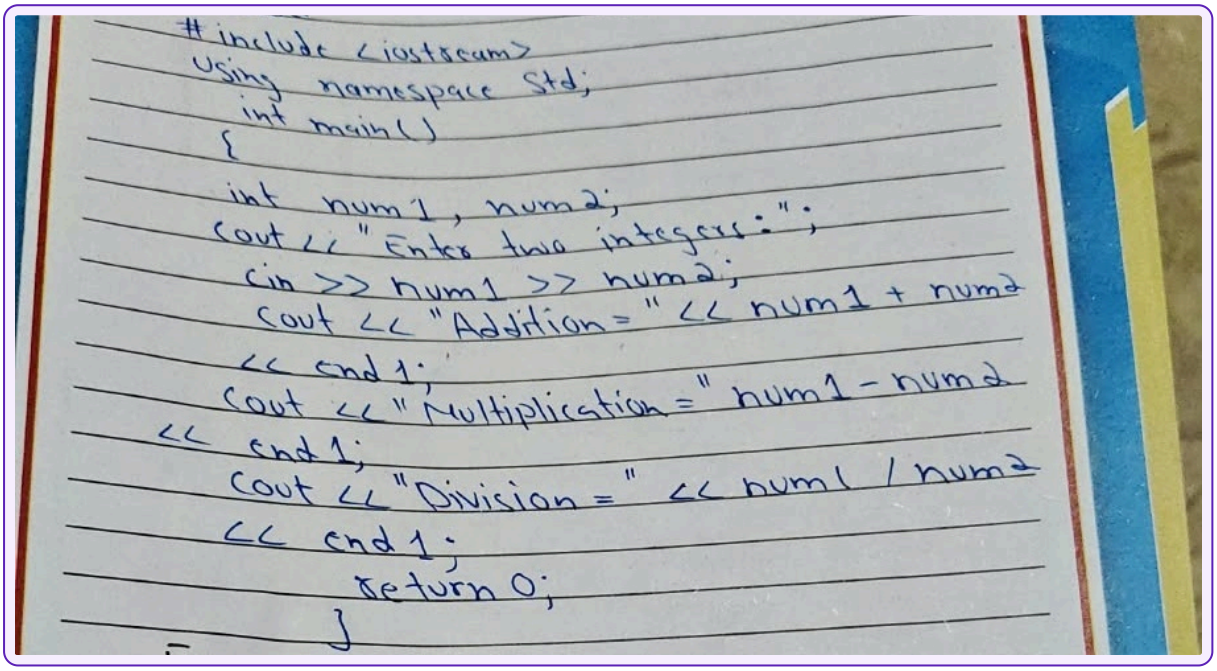
Compile Result

```
65
66
```

Lab Task 2:

Take two integers as input from the user, apply arithmetic operations on them (+, -, *, /) and print them on screen.

HANDWRITTEN CODE



```
#include <iostream>
using namespace std;
int main()
{
int num1, num2;
cout << "Enter two integers:";
cin >> num1 >> num2;
cout << "Addition = " << num1 + num2
<< endl;
cout << "Multiplication = " num1 - num2
<< endl;
cout << "Division = " << num1 / num2
<< endl;
return 0;
}
```

PHONE CODE

Coding C++

Auto saved at 12:20:57

RUN MENU

```cpp
#include <iostream>
using namespace std;
int main()
{
int num1, num2;
cout << "Enter two integers:";
cin >> num1 >> num2;
cout << "Addition = " << num1 + num2 << endl;
cout << "Multiplication = " << num1 * num2 << endl;
cout << "Division = " << num1 / num2 << endl;
cout << "Modulus = " << num1 % num2 << endl;
return 0;
}
```

PHONE OUTPUT

Compile Result

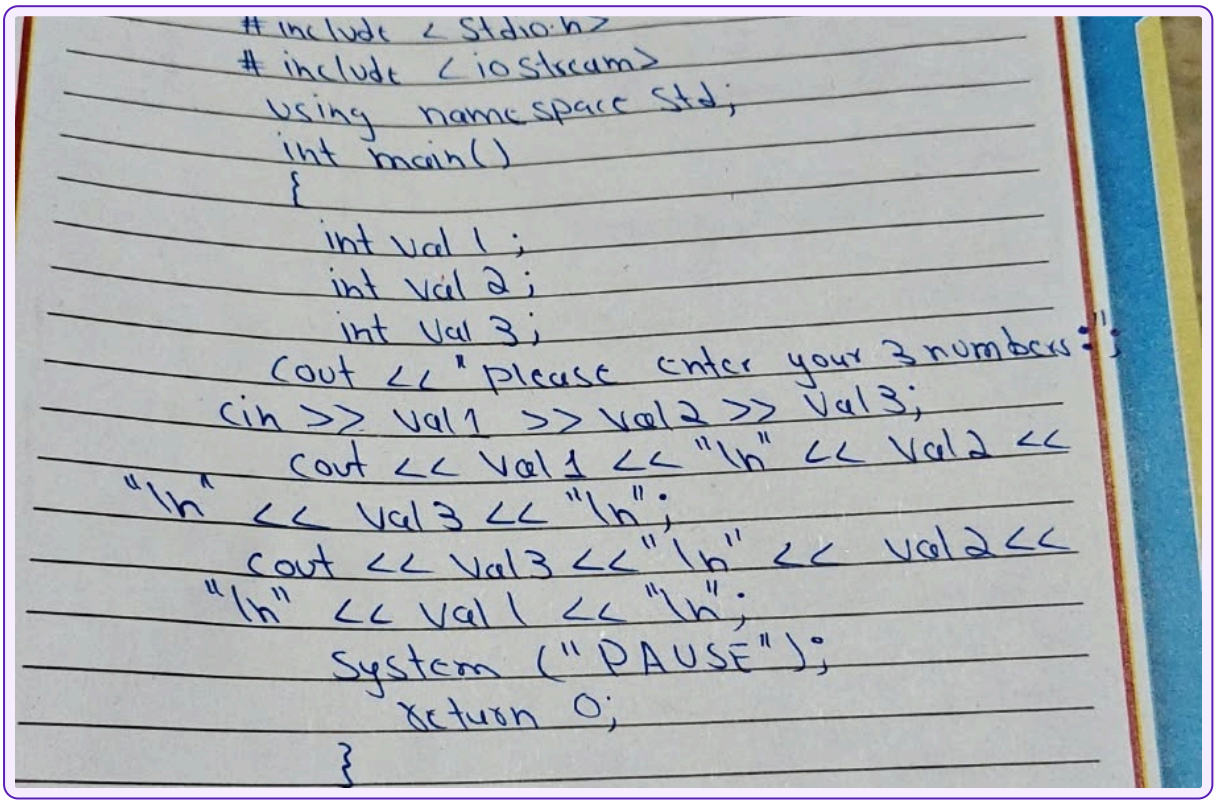
```
Enter two integers:10
3
Addition = 13
Multiplication = 30
Division = 3
Modulus = 1

[Process completed - press Enter]
```

Lab Task 3:

Write the output of the following code.

HANDWRITTEN CODE



```
#include <stdio.h>
#include <iostream>
using namespace std;
int main()
{
    int val 1;
    int val 2;
    int val 3;
    cout << "please enter your 3 numbers:";
    cin >> val1 >> val2 >> val3;
    cout << val1 << "\n" << val2 <<
"\n" << val3 << "\n";
    cout << val3 << "\n" << val2 <<
"\n" << val1 << "\n";
    System ("PAUSE");
    return 0;
}
```

PHONE CODE

Coding C++

Auto saved at 12:20:57

RUN MENU

```cpp
#include <iostream>
using namespace std;
int main()
{
int val1;
int val2;
int val3;
cout << "Please enter your 3 numbers:";
cin >> val1 >> val2 >> val3;
cout << val1 << "\n" << val2 << "\n" << val3 << "\n";
cout << val3 << "\n" << val2 << "\n" << val1 << "\n";
system("PAUSE");
return 0;
}
```

PHONE OUTPUT

Compile Result

```
Please enter your 3 numbers:3
5
7
3
5
7
7
5
3
sh: PAUSE: inaccessible or not found

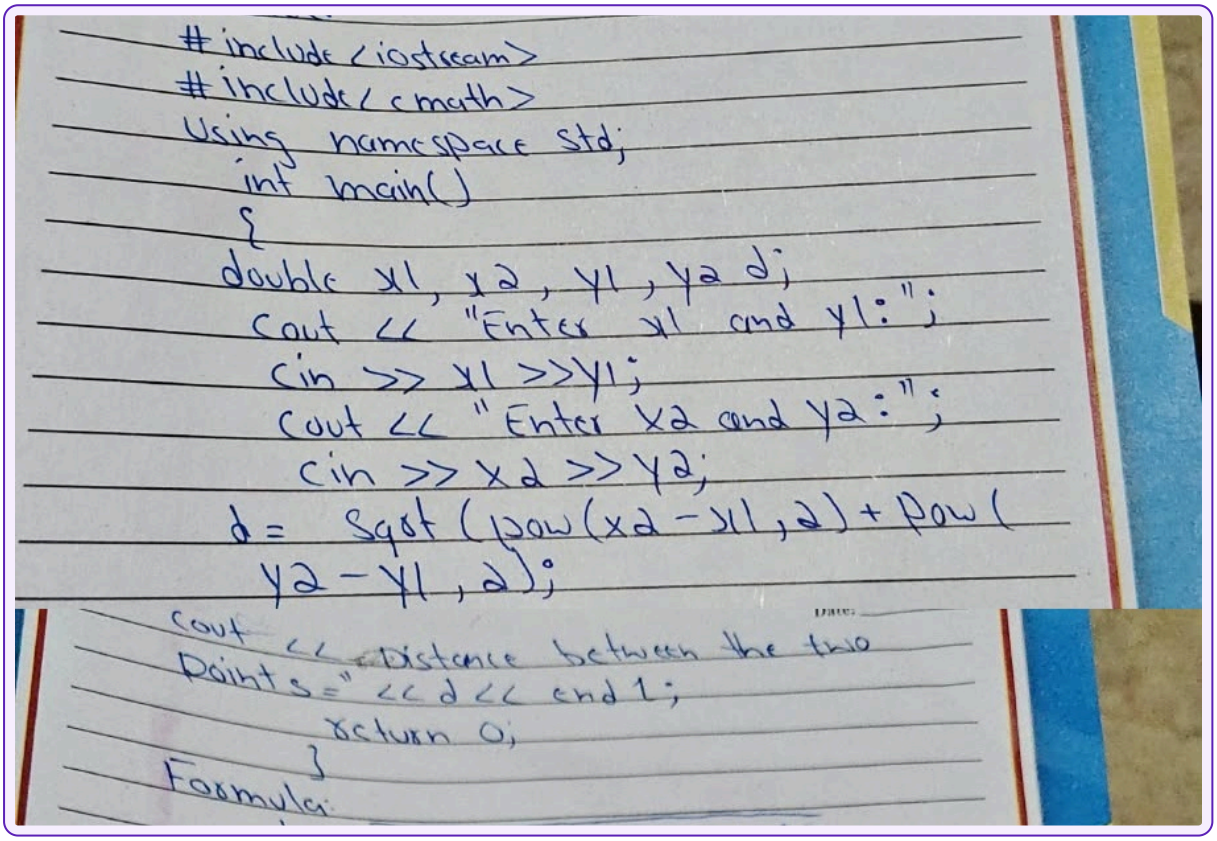
[Process completed - press Enter]
```

Lab Task 4:

Write a C++ program to calculate the distance between the two points.
Note: x1, y1, x2, y2 are all double values.

Formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

HANDWRITTEN CODE



PHONE CODE

Coding C++

Auto saved at 12:24:16

RUN MENU

```cpp
#include <iostream>
#include <cmath>
using namespace std;
int main ()
{
double x1, y1, x2, y2, d;
cout << " Enter x1 :";
cin >> x1;
cout << " Enter y1 :";
cin >> y1;
cout << "Enter x2 :";
cin >> x2;
cout << " Enter y2 :";
cin >> y2;
d = sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
cout << "Distance between two points =" << d << endl;
return 0;
}
```

PHONE OUTPUT

Compile Result

```
 Enter x1 :7
 Enter y1 :14
Enter x2 :8
 Enter y2 :16
Distance between two points =2.23607

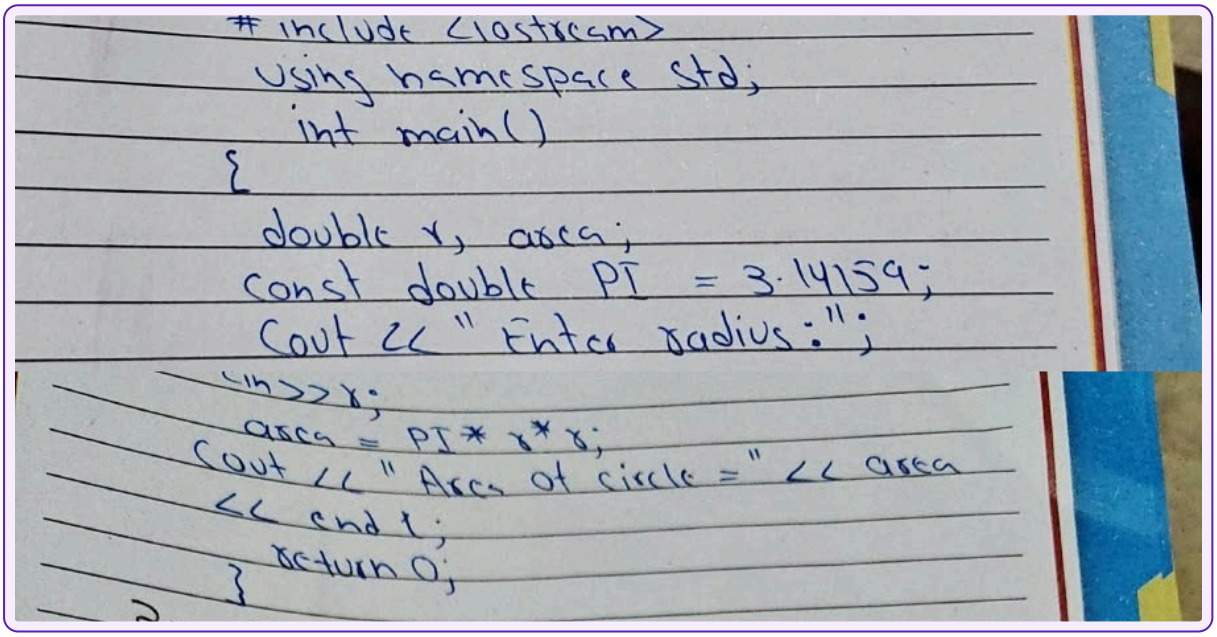
[Process completed - press Enter]
```

Lab Task 5:

1. Calculate the Area of a Circle (area = PI * r^2)
2. Calculate the Area of a Rectangle (area = length * width)
3. Calculate the Area of a Triangle (area = base * height * .5)

Part 1: Area of a Circle

HANDWRITTEN CODE



PHONE CODE

Coding C++

Auto saved at 15:01:25

RUN MENU

```
#include<iostream>
using namespace std;
int main()
{
float r, area;
cout<<"Enter radius :";
cin>>r;
area=3.14159*r*r;
cout<< "area of circle ="<< area;
return 0;
}
```

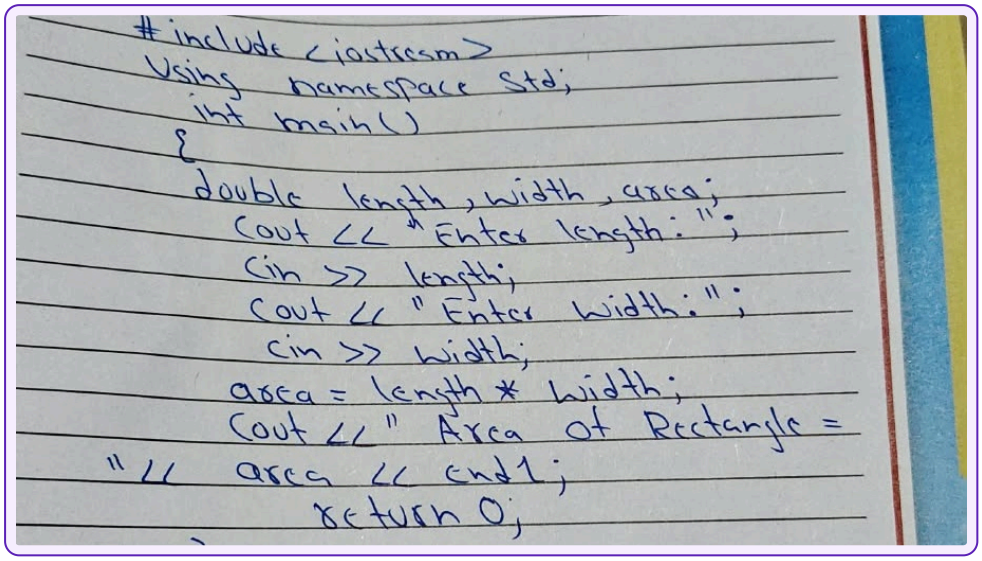
PHONE OUTPUT

Compile Result

```
Enter radius :287
area of circle =258770
[Process completed - press Enter]
```

Part 2: Area of a Rectangle

HANDWRITTEN CODE



PHONE CODE

Coding C++

Auto saved at 12:26:28

RUN MENU

```cpp
#include<iostream>
using namespace std;
int main()
{
double length, width, area;
cout<<"Enter length:";
cin>>length;
cout<<"Enter width:";
cin>>width;
area = length * width;
cout<<"Area of rectangle ="<< area << endl;
return 0;
}
```

PHONE OUTPUT

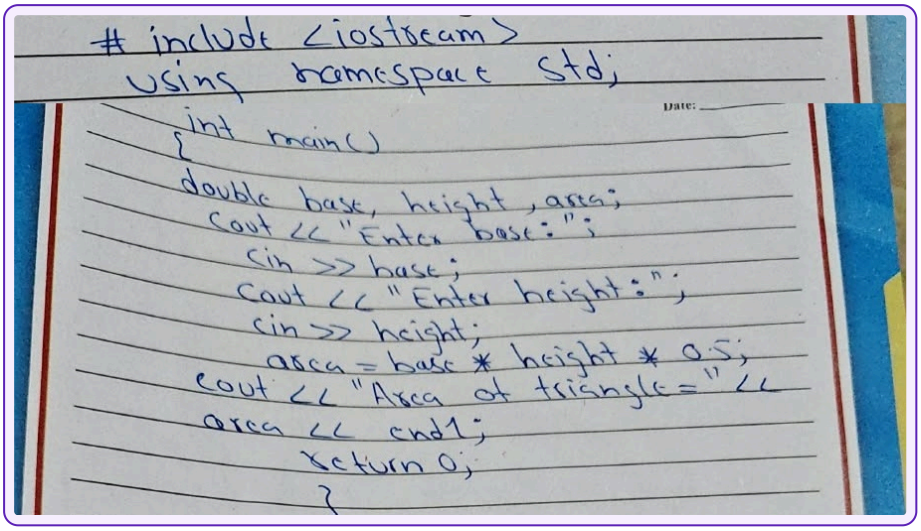
Compile Result

```
Enter length:50
Enter width:21
Area of rectangle =1050

[Process completed - press Enter]
```

Part 3: Area of a Triangle

HANDWRITTEN CODE



```
#include<iostream>
using namespace std;
int main()
{
double base, height, area;
cout<<"Enter base:";
cin>>base;
cout<<"Enter height:";
cin>>height;
area = base * height * 0.5;
cout<<"Area of triangle =";
area<<endl;
return 0;
}
```

PHONE CODE

Coding C++

Auto saved at 12:27:35

RUN MENU

```
1 #include<iostream>
2 using namespace std;
3 int main()
4 {
5 double base, height, area;
6 cout<<"Enter base:";
7 cin>>base;
8 cout<<"Enter height:";
9 cin>>height;
10 area = base * height*0.5;
11 cout<<"Area of triangle ="<< area << endl;
12 return 0;
13 }
```

PHONE OUTPUT

Compile Result

```
Enter base:8
Enter height:14
Area of triangle =56

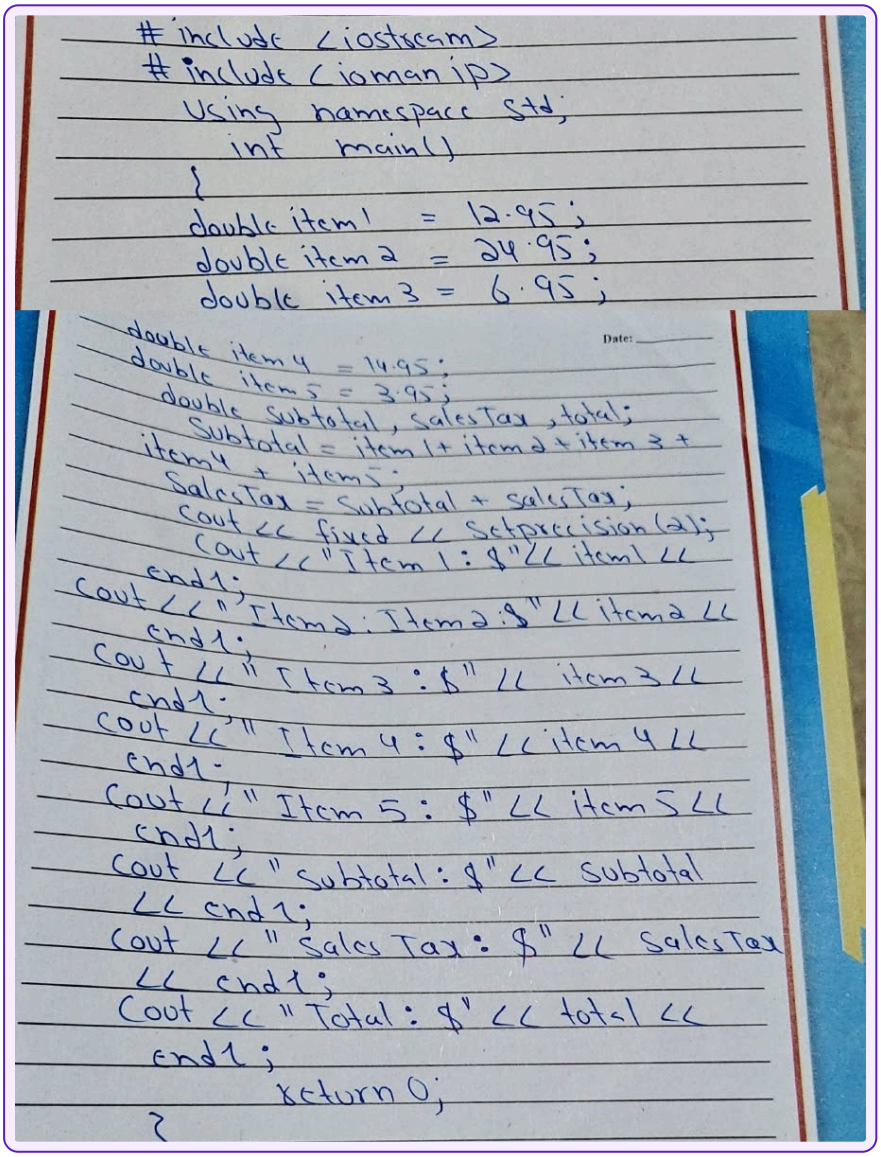
[Process completed - press Enter]
```

Lab Task 6:

Total Purchase: A customer in a store is purchasing five items. The prices of the five items are:
Price of item 1 = $12.95 | Price of item 2 = $24.95 | Price of item 3 = $6.95 | Price of item 4 = $14.95 | Price of item 5 = $3.95

Write a program that holds the prices of the five items in five variables. Display each item's price, the subtotal of the sale, the amount of sales tax, and the total. Assume the sales tax is 6%.

HANDWRITTEN CODE



```
#include <iostream>
#include <iomanip>
Using namespace std;
int main()
{
double item1 = 12.95;
double item2 = 24.95;
double item3 = 6.95;
double item4 = 14.95;
double item5 = 3.95;
double subtotal, salesTax, total;
Subtotal = item1 + item2 + item3 +
item4 + item5;
SalesTax = Subtotal * salesTax;
Cout << fixed << setprecision(2);
Cout << "Item 1: $" << item1 <<
endl;
Cout << "Item 2: Item 2: $" << item2 <<
endl;
Cout << "Item 3: $" << item3 <<
endl;
Cout << "Item 4: $" << item4 <<
endl;
Cout << "Item 5: $" << item5 <<
endl;
Cout << "subtotal: $" << Subtotal
<< endl;
Cout << "Sales Tax: $" << SalesTax
<< endl;
Cout << "Total: $" << total <<
endl;
return 0;
}
```

PHONE CODE

Coding C++

Auto saved at 15:03:10

RUN MENU

```cpp
#include<iostream>
using namespace std;
int main()
{
double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;

    double subtotal, tax, total;

    subtotal = item1 + item2 + item3 + item4 + item5;
    tax = subtotal * 0.06;
    total = subtotal + tax;

    cout << "Price of Item 1 = $" << item1 << endl;
    cout << "Price of Item 2 = $" << item2 << endl;
    cout << "Price of Item 3 = $" << item3 << endl;
    cout << "Price of Item 4 = $" << item4 << endl;
    cout << "Price of Item 5 = $" << item5 << endl;

    cout << "Subtotal = $" << subtotal << endl;
    cout << "Sales Tax = $" << tax << endl;
    cout << "Total = $" << total << endl;

    return 0;
}
```

PHONE OUTPUT

Compile Result

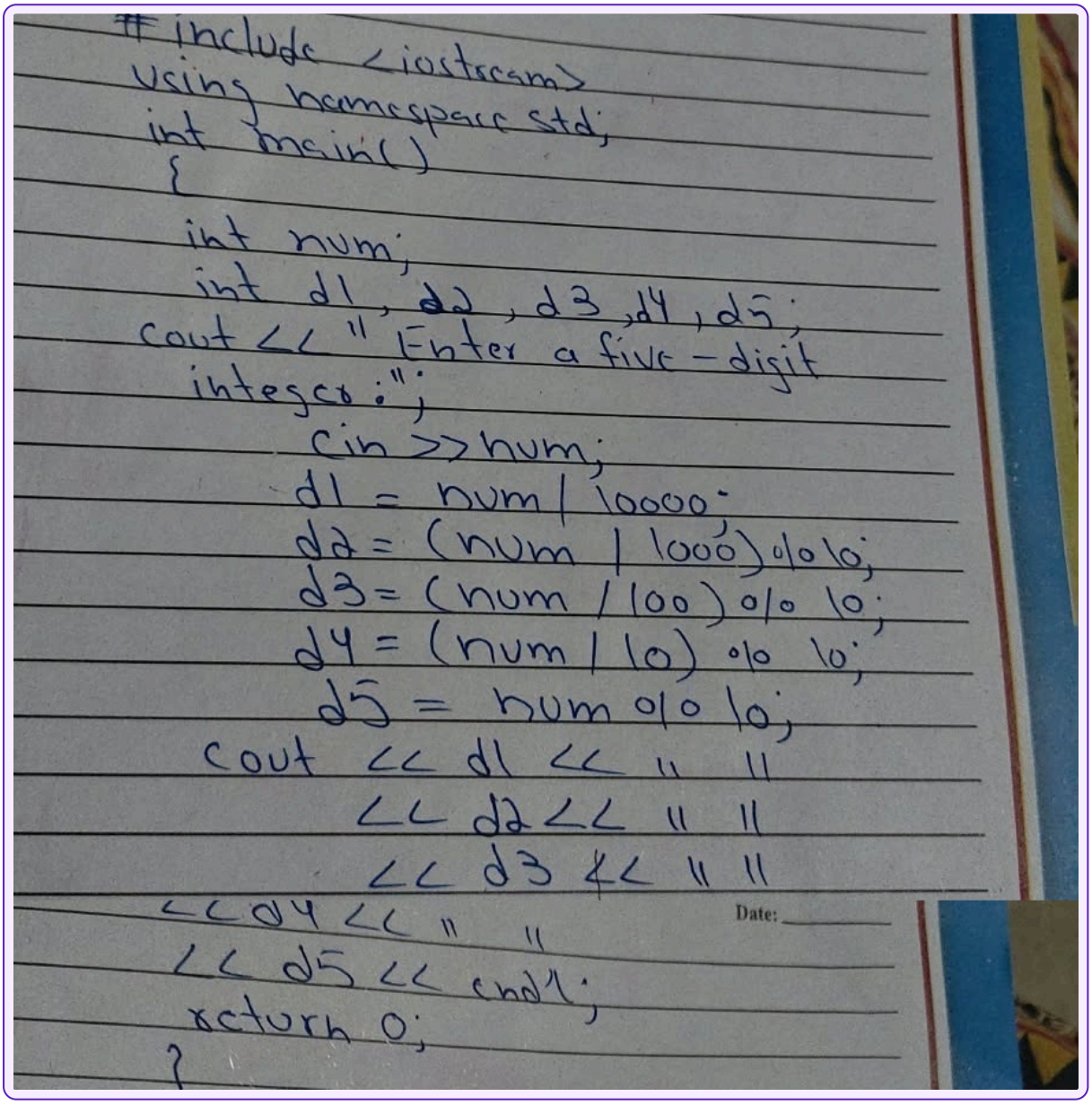
```
Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item 5 = $3.95
Subtotal = $63.75
Sales Tax = $3.825
Total = $67.575

[Process completed - press Enter]
```

Bonus Task:

Break Number: Write a program that inputs a five-digit integer, separates the integer into its individual digits and prints the digits separated from one another by three spaces each.

HANDWRITTEN CODE



```
#include <iostream>
using namespace std;
int main()
{
    int num;
    int d1, d2, d3, d4, d5;
cout << "Enter a five-digit
integer:";
    cin >> num;
    d1 = num / 10000;
    d2 = (num / 1000) % 10;
    d3 = (num / 100) % 10;
    d4 = (num / 10) % 10;
    d5 = num % 10;
cout << d1 << "  "
     << d2 << "  "
     << d3 << "  "
<< d4 << "  "
<< d5 << endl;
return 0;
}
```

PHONE CODE

Coding C++

Auto saved at 12:07:12

RUN MENU

```cpp
#include <iostream>
using namespace std;
int main()
{
int num;
int d1, d2, d3, d4, d5;
cout << "Enter a five-digit integer:";
cin >> num;
d1 = num / 10000;
d2 = (num / 1000) % 10;
d3 = (num / 100) % 10;
d4 = (num / 10) % 10;
d5 = num % 10;
cout << d1 << "    "
<< d2 << "    "
<< d3 << "    "
<< d4 << "    "
<< d5 << endl;
return 0;
}
```

PHONE OUTPUT

Compile Result

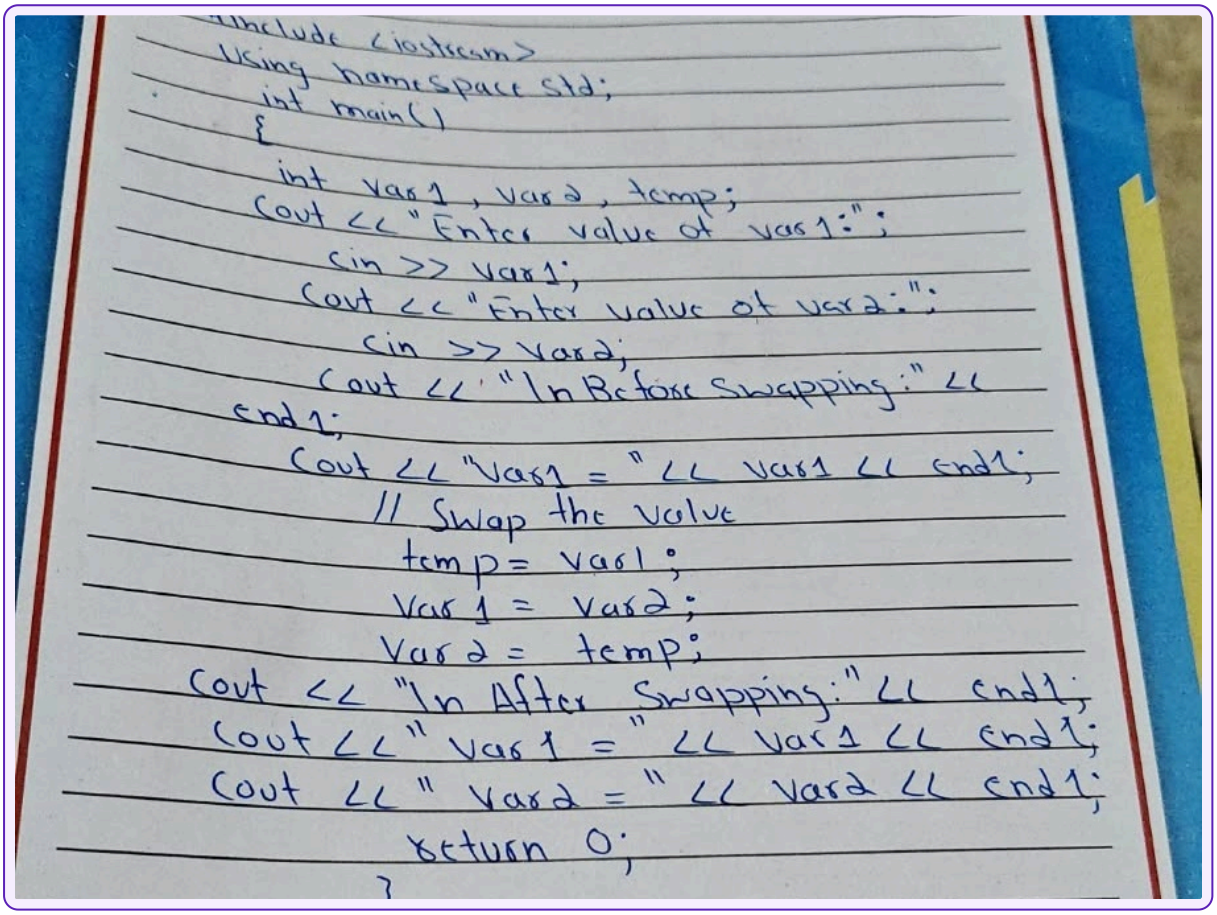
```
Enter a five-digit integer:98765
9    8    7    6    5

[Process completed - press Enter]
```

Home Task 1:

Exchange: Write a program that reads in two variables and swaps the value of these variables. For example, if a variable 'var1' contains 10 and variable 'var2' contains 20, after the swap operation the variable 'var1' contains 20 (value of var2) and variable 'var2' contains 10 (value of var1).

HANDWRITTEN CODE



```
#include <iostream>
Using namespace std;
int main()
{
  int var1, var2, temp;
  Cout << "Enter value of var1:";
    Cin >> var1;
  Cout << "Enter value of var2:";
    Cin >> var2;
  Cout << "\n Before Swapping:" <<
end1;
  Cout << "Var1 = " << var1 << end1;
    // Swap the value
      temp = var1;
      Var1 = var2;
      Var2 = temp;
Cout << "\n After Swapping:" << end1;
Cout << " Var1 = " << Var1 << end1;
Cout << " Var2 = " << Var2 << end1;
      return 0;
}
```

PHONE CODE

Coding C++

Auto saved at 15:05:10

RUN MENU

```
#include<iostream>
using namespace std;
int main()
{
int var1, var2, temp;

    cout << "Enter first value: ";
    cin >> var1;

    cout << "Enter second value: ";
    cin >> var2;

    temp = var1;
    var1 = var2;
    var2 = temp;

    cout << "After swapping:" << endl;
    cout << "var1 = " << var1 << endl;
    cout << "var2 = " << var2 << endl;

    return 0;
}
```

PHONE OUTPUT

Compile Result

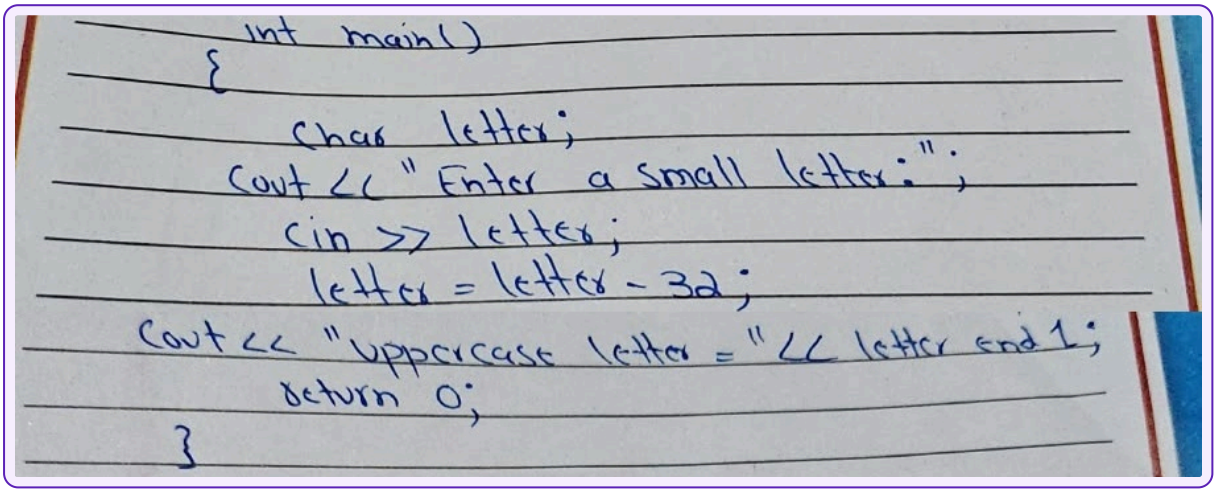
```
Enter first value: 32
Enter second value: 15
After swapping:
var1 = 15
var2 = 32

[Process completed - press Enter]
```

Home Task 2:

Make Upper: Write a program that prompts the user to enter an alphabet in small caps (a, b, c, … z) and display the entered alphabet into its upper caps (A, B, C, … Z). (Hint: ASCII)

HANDWRITTEN CODE



```
int main()
{
    char letter;
cout << "Enter a small letter:";
    cin >> letter;
    letter = letter - 32;
cout << "uppercase letter =" << letter endl;
    return 0;
}
```

PHONE CODE

Coding C++
Auto saved at 15:06:53
RUN MENU

```cpp
#include<iostream>
using namespace std;
int main()
{
int var1, var2, temp;

    char ch;

    cout << "Enter a small alphabet: ";
    cin >> ch;

    ch = ch - 32;

    cout << "Uppercase alphabet = " << ch << endl;

    return 0;

}
```

PHONE OUTPUT

Compile Result

```
Enter a small alphabet: n
Uppercase alphabet = N

[Process completed - press Enter]
```

End of Assignment

PROGRAMMING FUNDAMENTALS · LAB 2